



CHISHOLM CATHOLIC COLLEGE
Year 11 Mathematics Methods
Test 1 2025

Total Mark

/44

Section 1: Calculator - free

Name:

Time: 25 minutes maximum

Solutions

Teacher:

Marks available: 20

Instructions:

- Answer all questions in the spaces provided.
- Calculators and notes are not permitted.
- Formula sheet provided.
- Where questions or parts of questions are worth more than 2 marks, justification is required.

Question 1**[2 marks]**

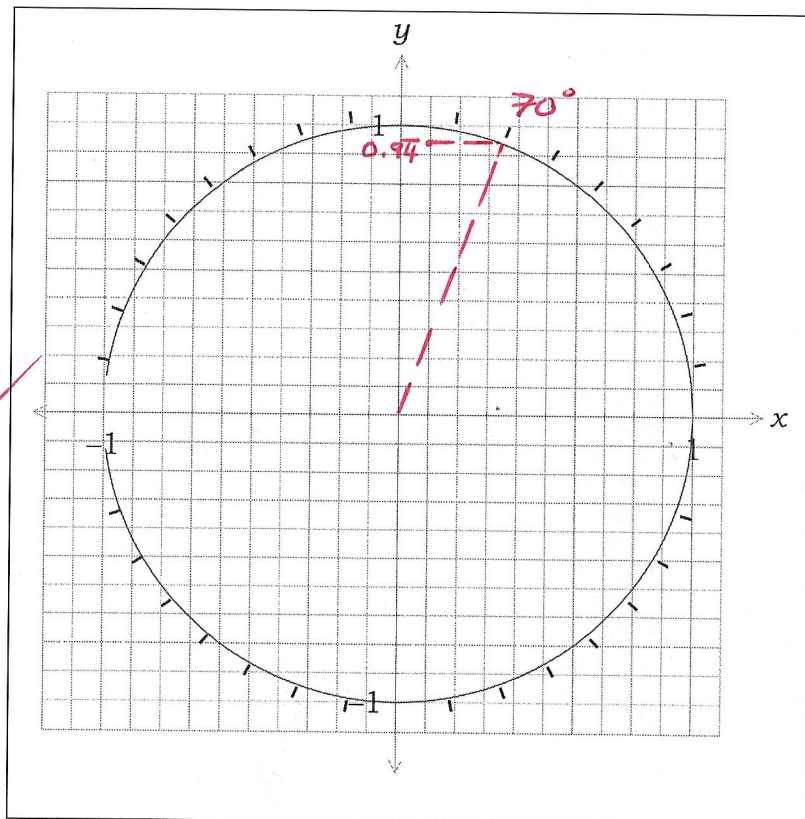
- a) Use the unit circle that has increments of 10° below to determine the value of (to 2 decimal places if necessary):

(i) $\sin 70^\circ = 0.94$ ✓

Accept 0.93 - 0.95

(ii) $\cos \frac{3\pi}{2}$

$\cos 270^\circ = 0$ ✓



Question 2**[1, 1, 2 = 4 marks]**

- a) Write 225° as an angle in radians in simplest form

$$\frac{225\pi}{180} = \frac{45\pi}{36} = \frac{5\pi}{4} \quad \checkmark$$

- b) Convert $\frac{7\pi}{3}$ to degrees

$$\frac{7\pi}{3} \times \frac{180}{\pi} = 420^\circ \quad \checkmark$$

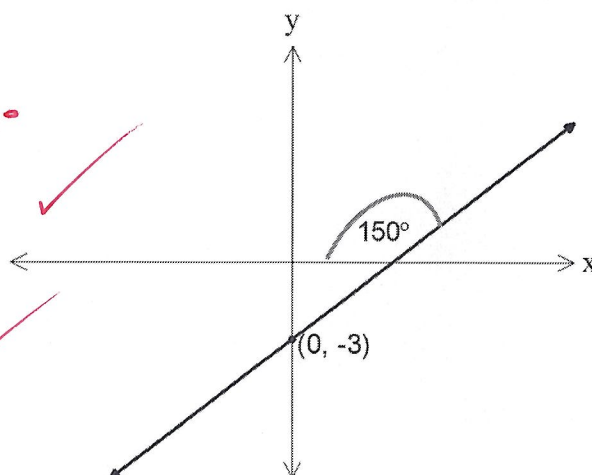
- c) Write down the angle of inclination of the line below and hence determine the gradient of the line as an exact value.

Angle of inclination = 30° ✓

Gradient = $\tan 30^\circ$ ✓

$$= \frac{1}{\sqrt{3}}$$

$$= \frac{\sqrt{3}}{3} \quad \checkmark$$

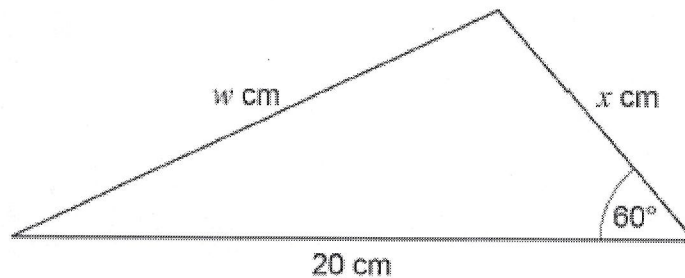


Question 3

[2, 3, 2 = 7 marks]

The area of this triangle is $25\sqrt{3}$

Diagram not to scale



- a) Show that the length of x is 5 cm

$$25\sqrt{3} = \frac{1}{2} \times x \times 20 \times \sin 60^\circ \quad \checkmark$$

$$25\sqrt{3} = 10x \times \frac{\sqrt{3}}{2}$$

$$25\sqrt{3} = 5x\sqrt{3} \quad \checkmark$$

$$x = 5$$

- b) Find the value of w . Give your answer in the form $a\sqrt{b}$ where a and b are integers greater than 1

$$w^2 = 5^2 + 20^2 - 2(5)(20)\cos 60^\circ \quad \checkmark$$

$$w^2 = 425 - 200 \times \frac{1}{2}$$

$$w^2 = 325$$

$$w = \sqrt{325} \quad \checkmark$$

$$w = \sqrt{25} \times \sqrt{13} \quad \checkmark$$

$$w = 5\sqrt{13}$$

- ~~c) Show that the value of $\cos 30^\circ \times \tan 60^\circ + \sin 30^\circ$ is an integer.~~

$$= \frac{\sqrt{3}}{2} \times \sqrt{3} + \frac{1}{2}$$

$$= \frac{3}{2} + \frac{1}{2} \quad \checkmark$$

$$= \frac{4}{2} \quad \checkmark$$

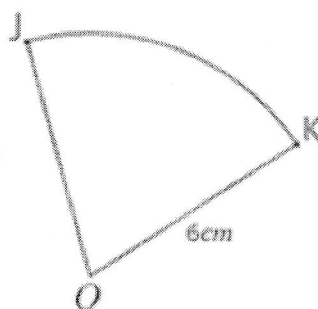
$$= 2$$

(7)

Question 4

³
[4 marks]

In the diagram, O is the centre of the circle with radius 6cm . Arc JK has a length measuring 2π cm. Determine the exact area of sector OJK .



$$\theta = \frac{2\pi}{6}$$

$$\theta = \frac{\pi}{3} \quad \checkmark$$

$$\text{Area} = \frac{1}{2} \times 6^2 \times \frac{\pi}{3} \quad \checkmark$$

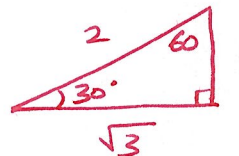
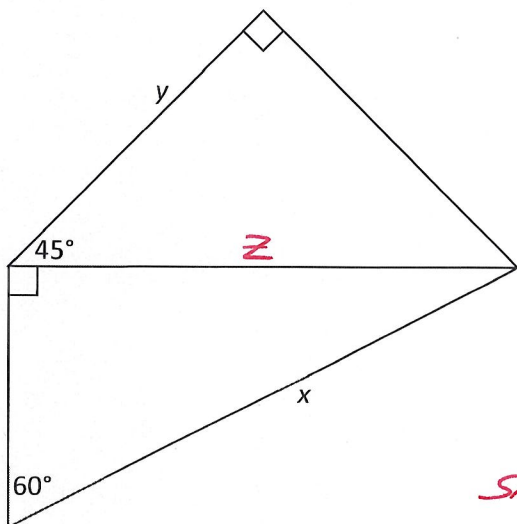
$$\text{Area} = 6\pi \text{ cm}^2 \quad \checkmark$$

(3)

Question 5

[4 marks]

Using the diagram below show that $y = \frac{\sqrt{3}x}{2\sqrt{2}}$



$$\sin 60^\circ = \frac{z}{x}$$

$$\therefore z = x \cdot \sin 60^\circ$$

$$= \frac{\sqrt{3}}{2} x$$

$$\cos 45^\circ = \frac{y}{z}$$

$$\frac{1}{\sqrt{2}} = \frac{y}{z}$$

$$y = \frac{z}{\sqrt{2}}$$

$$= \frac{\frac{\sqrt{3}}{2} x}{\sqrt{2}}$$

$$= \frac{\sqrt{3}x}{2\sqrt{2}}$$

accept $y = \frac{\sqrt{6}x}{4}$

End of Section 1

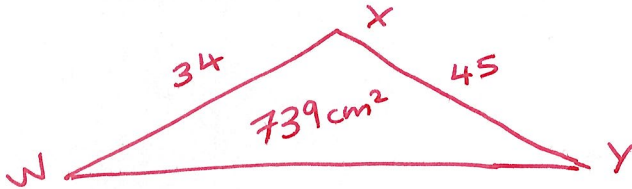
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Question 6

[1, 2, 2, 2 = 7 marks]

A triangle WXY has $w = 45$ cm, $y = 34$ cm and an area of 739 cm².

(a) Sketch a triangle to show this information.



(b) Determine the size of $\angle X$.

$$739 = \frac{1}{2} (34)(45) \sin X \quad \checkmark$$

$$X = 75^\circ \text{ or } \underline{105^\circ} \quad \checkmark$$

(c) Show that $x \approx 63$ cm.

$$x^2 = 34^2 + 45^2 - 2(34)(45) \cos 105^\circ \quad \checkmark$$

$$x = 63.0316 \dots$$

$$x \approx 63 \text{ cm} \quad \checkmark$$

(d) Show that $\angle Y \approx 31^\circ$.

$$\cos Y = \frac{63^2 + 45^2 - 34^2}{2(63)(45)} \quad \checkmark$$

$$Y = 31.43 \dots$$

$$Y \approx 31^\circ \quad \checkmark$$

OR

$$\frac{\sin Y}{34} = \frac{\sin 105^\circ}{63}$$

(7)

Question 7**[2 marks]**

The sides adjacent to the right-angle in a right angled triangle have lengths 65 cm and 72 cm.

If the smallest angle in the triangle is α , determine an exact value for

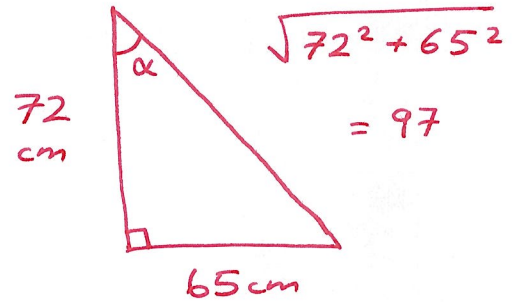
(i) $\tan \alpha$.

$$\frac{65}{72}$$



(ii) $\sin(90^\circ - \alpha)$.

$$\frac{72}{97}$$

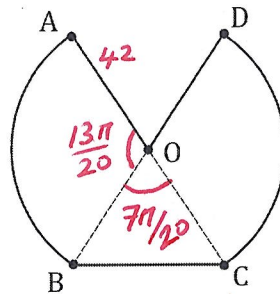


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Question 8

[4, 3 = 7 marks]

In shape $OABCD$ below, $\angle AOB = \frac{13\pi}{20}$ and AC, BD are diameters of the circle with centre O and radius 42 cm.



(a) Calculate the perimeter of $OABCD$.

$$\begin{aligned}\angle BOC &= \pi - \frac{13\pi}{20} \\ &= \frac{7\pi}{20}\end{aligned}$$

$$BC^2 = 42^2 + 42^2 - 2(42)(42) \cdot \cos \frac{7\pi}{20}$$

$$BC = 43.89 \text{ cm}$$

$$\text{Arc } AB = 42 \times \frac{13\pi}{20}$$

$$\approx 85.77 \text{ cm}$$

$$\text{Perimeter} = 42 + 42 + 43.89 + 2 \times 85.77 = 299.43 \text{ cm}$$

(b) Calculate the area of $OABCD$.

$\triangle BOC$

$$\begin{aligned}\text{Area} &= \frac{1}{2} \times 42 \times 42 \times \sin \frac{7\pi}{20} \\ &= 785.87 \text{ cm}^2\end{aligned}$$

$\triangle AOB$

$$\begin{aligned}\text{Area} &= \frac{1}{2} \times 42^2 \times \frac{13\pi}{20} \\ &= 1801.08 \text{ cm}^2\end{aligned}$$

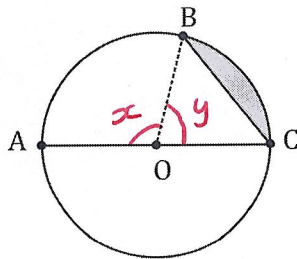
$$\begin{aligned}\text{Total Area} &= 785.87 + 2 \times 1801.08 \\ &= 4388.03 \text{ cm}^2\end{aligned}$$

(7)

Question 9

[4, 4 = 8 marks]

- (a) The circle shown has centre O and diameter AC of length 50 cm. Determine the shaded area given that $2 \times \angle AOB = 3 \times \angle BOC$.



- i) Show that $\angle BOC = \frac{2\pi}{5}$

$$x + y = \pi$$

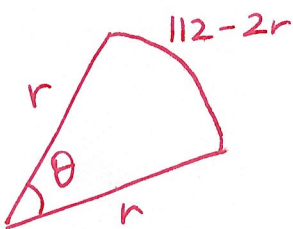
$$2x = 3y$$

By Classpad Simultaneous $y = \frac{2\pi}{5}$ ($\angle BOC$)

- ii) Determine the shaded area

$$\begin{aligned} \text{Segment Area} &= \frac{1}{2} \times 25^2 \left(\frac{2\pi}{5} - \sin \frac{2\pi}{5} \right) \\ &= 95.49 \text{ cm}^2 \end{aligned}$$

- (b) A sector of a circle has a perimeter of 112 cm and an area of 735 cm². Determine the radius of the circle.



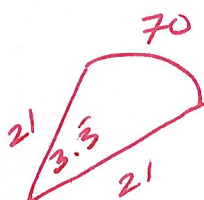
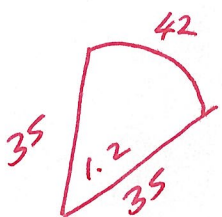
From perimeter $\theta = \frac{112 - 2r}{r}$

From area

$$735 = \frac{1}{2} r^2 \theta$$

By Classpad $\rightarrow$ Simultaneously

$$r = 35 \text{ cm} \quad \text{or} \quad r = 21 \text{ cm}$$



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